Manufacturer product index: what we built, what we learned, what we propose

Team packet · October 2026 · Everything from the rulebook and answer keys through the current scraper, the Approach B experiments and the proposals. Percentages are against the answer keys unless stated.

1. Summary

The goal. Turn manufacturer websites into a clean product index that is about 97% accurate and works for any manufacturer without custom code. The index has one record per product family, holding its documents, CAD files, images, specs, sizes/part numbers and prices.

What we did, in order:

1. **Wrote the rulebook (SPEC v1)**: what counts as a family, a document, a good image, a spec value or a price.
2. **Built answer keys for 4 manufacturers** (Bray, Crane, Cimberio, Sioux Chief) and a scorer, so every version is measured the same way.
3. **Built a new scraper** (the "current scraper") and improved it over 10 rounds. It then failed the test on sites it had never seen.
4. **Ran a gap analysis** against the old agent's original data. Most misses were items we had seen and then filtered out.
5. **Designed "collect wide, sort smart"** and tested it (Approach B). The old agent's link list goes in, and a new sorting step produces the index. We ran 10 experiments on it, including an AI helper.
6. **Built the site-profile step**, which reads how each manufacturer groups its own products. On Sioux Chief it cut families from 4,574 to 603 (700 real ones), but a few merged cards still hurt document accuracy (section 10a).

Old agent's list already contains	Answer-key double-check agreement	Sioux Chief families found	Specs correct, any version
98–100%	**82–100%**	**54% → 90%**	**15–67%**
of key families, docs, CAD and images on 3 of 4 sites	two labellers, working blind	across our experiments (current scraper 93%)	the biggest remaining gap

The five things we learned:

1. **Finding links is mostly solved; sorting them is the problem.** The old agent finds nearly everything. Every tool loses its points afterwards: deciding what a page is, grouping part numbers into families, choosing files and reading specs.
2. **Fixed rules break on new sites.** Rules such as "more than 15 links = category page" or "assets on other domains are junk" worked on the two sites we tuned on, then failed on the next ones.
3. **Every manufacturer already says what its families are**, through family pages, category cards or size tables. We should read that structure, not guess it from part numbers.
4. **AI helps only where it's aimed well.** Asking it "are these two pages the same family?" was 58% right and couldn't scale. Reading spec tables and settling conflicts, with a verified quote, is the better use.
5. **Tuning on a few sites flatters you.** Ten rounds of gains on 2 sites didn't carry over to new ones. Only scoring on unseen manufacturers proves "works for all".

What we propose (section 10):

1. A "how is this site organised?" step, run on every site. **Built**; first results in 10a. Next: a shared shape catalogue, so new sites need settings only, not new rules.
2. A few free extra fields from the old agent, "found on" and page copies first.
3. Better spec reading, with AI backed by quotes.
4. Answer keys for 3–5 new manufacturers, to test blind.

Later, the database tables in DATA_MODEL.md.

2. The starting point: the old agent's output

The old agent crawls a manufacturer's site and writes a CSV with two columns: `url` and `url_type`. We have its output files only, not its code.

Site	Rows in its list	Families in list	Main docs in list	CAD in list	Images in list	Notes
Bray	2,211	100%	100%	100%	98%	Only 10% of its page rows are product families
Crane	1,944	99%	100%	67%	100%	78% of page rows are families; CAD labels wrong
Cimberio	1,181 pages	100%	54%	0%	100%	Price list and half the main docs never listed; 1,149 page rows on an off-site domain
Sioux Chief	15,017	100%	100%	100%	100%	One page per part number (6,700+ pages for 700 families)

"In list" means the item appears anywhere in the CSV. It doesn't mean the agent knows which product the item belongs to.

What it does well: discovery. Its recall is higher than any crawler we built. It also reached Dixon and Apollo, two sites that block our requests completely.

What it lacks:

- **No relationships.** It doesn't record which page a link was found on, so we can't tell which product a PDF or image belongs to without crawling again.
- **No sorting.** There are no families, part numbers, specs or prices, and a lot of noise: on Bray, 9 in 10 page rows aren't products.
- **Inconsistent labels.** The same kind of file gets different `url_type` labels on different sites, for example CAD viewers labelled as "document".
- **Some discovery gaps.** Cimberio's price list and about half its main documents are missing.

3. The rulebook (SPEC v1)

Before measuring anything, we wrote down exactly what a correct answer looks like. It was locked on 9 Oct 2026 and clarified 11 times during the work, each time when two labellers disagreed.

Topic	Rule in short
Unit of record	One record per **product family**, with one canonical URL. A **variant** is an orderable item with its own part number. Option codes (e.g. ".AT", suffixes) are not variants.
Crawling	Polite: about 1 request per second per host, honest user agent, no bot-protection workarounds. Pages that robots.txt disallows are in scope.
Language	US/CA English first, then generic English. International English is kept but flagged. Multilingual docs only count if English comes first and is at least 50% of the text.
Documents	All PDFs, labelled by type (spec sheet, submittal, IOM manual, brochure, price list…). The type is decided by **content**, not file name.
Images	Product photos and dimensional drawings only. Keep the largest version (at least 300 px), with no duplicates, infographics or installation/lifestyle photos.
CAD	Direct files are recorded. Hosted or gated CAD libraries are recorded as links only; no forms are filled in.
Specs	13 fields (materials, end connections, pressure, temperature, sizes, standards, certifications, applications…), using controlled vocabularies. Page and spec sheet are cross-checked as agree, refines or conflict. Conflicts are kept, never silently resolved.
Evidence	Every spec value cites a **verbatim quote** from a stored copy of its source, and the validator checks the quote.
Price	From manufacturer sources only, per variant. Availability is always filled: price shown, login required, request a quote, or not published.

4. Answer keys and the scorer

How each key was built (one labelling session per site, with a checkpoint for the owner's corrections):

1. **A complete family list** from the site's sitemap and its category navigation, using a written grouping rule. Family counts: Bray 103, Crane 217, Cimberio 32, Sioux Chief 700.
2. **15 families per site labelled in depth**, picked at random (seed 42): every document, image, CAD file, variant, price and spec, each backed by a stored quote.
3. **An ambiguity log** recording every judgement call and its reason.
4. **A blind second labeller** re-labelled 6 families per key without seeing the first key. Disagreements were resolved by clarifying the rulebook.

Key	Blind agreement, before → after rulebook fixes	Main sources of disagreement
Crane	specs 78% → 95%; documents 20/29 → 29/29; images → 17/17	What "standards" covers; image URLs that no page links to
Cimberio	specs 81% → 82%; variants 27/27; images 9/9; CAD 2/2	The blind labeller missed a public price list; material wording
Bray (images + doc types)	doc types 73% → 100%; images 76% → 88%	Led to the "classify by content" and image edge-case rules
Sioux Chief	vocabulary pass only	Its family grouping was approved by the owner at the checkpoint

Limits to keep in mind:

- Docs, images, specs and prices are measured on 15 families per site. Family counts use the full list.

- Sioux Chief's 700 is the key's catalog model: one family per category card, split where a card mixes product lines (28 times). Spare-parts lines count as separate families. Only the 15 deep families were checked one by one.
- Apollo and Dixon block automated access, so they have no key. Milwaukee and Caleffi are kept untouched as a future blind test and have no keys yet.

The scorer checks every output file against the rulebook: schema, quotes, wrong-language and duplicate checks. It then scores each measure against the key: families found and correct, duplicates, docs, CAD, images (including best version), variants, specs (accuracy and completeness, by field), conflicts, prices. It can also score the old agent's CSV, but only for "is it in the list?". It never fetches anything, so scores are repeatable.

5. The current scraper: built, tuned, then tested blind

A crawler plus extractor written to the rulebook. It finds pages from the sitemap and menus, groups them into families, then reads tables and PDFs. It contains no site-specific code, and a test enforces that.

10 improvement rounds on the two development sites (Bray, Crane), average of both:

Measure	Round 1	Round 10	Main changes
Main docs found	81%	96%	Allow up to 30 PDFs per family instead of 10
Part numbers (variants) found	0%	87%	Read model/part-number tables, model headings and identifier lists
Price status correct	0%	89%	Every family reports a price status
Specs correct	54%	64%	Temperature formats, sizes from DN/NPS columns, applications from prose
Spec completeness	74%	84%	
Families found / correct, images	99 / 91 / 79	unchanged	

Then the blind test, run once on sites it had never seen:

Measure	Sioux Chief: scraper	Sioux Chief: old agent list	Cimberio: scraper	Cimberio: old agent list
Families found	92%	100%	97%	100%
Families produced (key)	6,270 (700)	–	47 (32)	–
Main docs found	46%	100%	25%	54%
CAD found	14%	100%	–	0%
Images found	14%	100%	0%	100%
Part numbers found	3%	–	0%	–
Specs correct	15%	–	35%	–
Run time	~9 h	–	2 min	–

Why it failed:

1. **Sioux Chief has one page per part number.** The scraper treated each page as a separate family (6,270 vs 700), which scattered docs, images and sizes across them.
2. **Cimberio stores its images and most datasheets on its store/CDN domains.** The scraper refused to follow those links.
3. **Shared datasheets were discarded as "site chrome"**, even when their text named the product.
4. **The image filter was too strict** (about 79% found even on the development sites).
5. **It was slow:** fetching one file at a time and parsing the same PDFs again and again.

We fixed the speed part afterwards. Each PDF is now parsed once, keyed by its content fingerprint: on Bray a run went from 13 min to 7 min, and to 1 min 40 s on a re-run, with byte-identical output.

6. Gap analysis: what we missed compared with the original data

For every document, image and CAD file in the keys, we asked three questions: did the old agent list it, did our crawl see it, and why did we miss it?

Site	Key items	We found	Missed: seen, then filtered out	Missed: fetched, then dropped	Missed: never seen
Bray	164	153	6	3	2
Crane	136	113	21	2	0
Cimberio	56	10	39	0	7
Total	356	276	**66 (83% of misses)**	5	9

What it means: we didn't need a better crawler. We needed to stop throwing things away. The old agent had 9 items that our crawl never saw. Everything else we missed, we had already seen and filtered out. That led to the new design.

7. The new design: "collect wide, sort smart"

The old agent *collects too much and doesn't sort it*. The current scraper *sorts well but filters too hard*. The new design combines their strengths:

Layer	Job	Who
1. Collect	Every page, PDF, image and CAD link, on any host (CDNs included), with where each was found.	The old agent (decided). Today it supplies bare CSVs, later the extra fields in section 10b.
2. Sort with rules	Group pages into families; drop obvious junk; keep obvious matches; read easy fields. Every item gets a **keep / drop / unsure** decision with a reason, and nothing is dropped silently.	The new processing step (Approach B), deterministic
3. AI for unsure items only	"Does this file belong to product X?", free-text specs, page-vs-PDF conflicts. The AI must quote the source text exactly, and the quote is checked.	Resolver, with cached answers and a budget limit
Output + audit	One record per family in the existing format, plus a list of everything rejected and why.	Scored by the same harness

Self-healing loop (designed, not yet tested). The AI answers unsure items. When it answers many similar items the same way, the pattern becomes a reusable **data rule**: a spec-name synonym, a section rule or a per-site profile entry. Cases a data rule can't express go to a coding agent as a reviewed PR. Over time there are fewer unsure items and AI cost falls.

Guardrails:

- Answer keys only measure, never teach.
- No quote, no answer.
- Every rule is versioned and can be rolled back.
- A new rule can't silently change a confident decision.
- Held-out sites stay untouched.
- Cost is bounded per site.

8. Approach B: what we built and what each experiment showed

Built:

- A `process` command that takes the old agent's link file and writes the index in the same format as the current scraper.
- A decisions log: every keep, drop or unsure decision, with its reason and evidence.
- Blocked-site tracking: after 8 blocked replies in a row, it stops asking that host.
- Fetching that works on several hosts at once while staying polite to each.
- A reader for today's CSV (L0) and for the richer format we propose (L1), with a validator.
- An L1 simulator, and a resolver (AI helper) with quote checking, caching and a budget.

Rule for every experiment (the **keep rule**): the average score must rise, and no key measure on any site may drop more than 2 points.

Every experiment, in order

Experiment	What changed	Result	Decision
First version	A new step that sorts the old agent's link list into families, documents, images and specs.	Better grouping than the current scraper on Bray and Cimberio. On Sioux Chief, 46% of families were lost to the "listing page" rule.	Starting point
Round 1	Match part numbers as whole words; a performance chart is not a product photo.	Average score 72.1 → 74.1. Cimberio image accuracy 15% → 50%. Bray images found fell 5 points.	Kept
Round 2	Keep shared documents linked from the product's own page.	Average 74.1 → 74.4. Bray main docs 85% → 93%, other docs 81% → 95%.	Kept
Round 3	Keep documents that cover several products.	Crane other docs 55% → 94%, but specs got worse. Average 74.4 → 73.8.	Rejected, undone
Round 4	Keep an image named by the family's own series code.	Average 74.4 → 74.6. Crane images found 85% → 93%.	Kept
Crane fix	Stop reading "931" in DA931 as a size, which had merged different products.	Crane 208 → 218 families (key: 217). Scores unchanged.	Kept
Richer link list (L1)	Pretend the old agent also saved page copies and file details.	Same accuracy, 19% fewer downloads, Sioux Chief re-run 49 → 36 min. Image accuracy fell (no duplicate check without downloading).	Information only
Listing-rule fix (Option A)	Don't drop a "listing-looking" page if its web address and content say it's a product; flag it as unsure instead.	Sioux Chief families found 54% → 88%, other docs 33% → 87%. Other sites unchanged.	Kept

Experiment	What changed	Result	Decision
Item-table grouping rule	Merge two part-number pages when each lists the other in its item table.	Right 99.6% of the time. Sioux Chief 5,725 → 4,574 families, sizes found 4% → 19%, images 27% → 40%. Main docs and CAD fell because of how one split family is scored.	In the PR (formally failed the keep rule)
Claude helper	Ask Claude "are these two pages the same family?" for pairs the rules couldn't settle.	323 of 497 answered before the usage limit. Only 58% right when it said "same". 4,574 → 4,515 families.	Stopped: not the right lever

Targets we set for this phase, and where we are

Target	Result
Image and document recall ≥ 95%	Not met. Main docs: Bray 93, Crane 100, Cimberio 54 (the list itself is missing docs), Sioux Chief 39–43. Images: Bray 73, Crane 93, Cimberio 100, Sioux Chief 40.
Precision drop ≤ 2 points vs current scraper	Met on Bray, Crane, Cimberio. Not met on Sioux Chief images (62 vs 79 correct).
~6,000-product site in under 4 h	Not met on bare CSVs (4 h 09 min). Estimated about 2 h with page copies from the old agent. Re-runs from saved copies take 36–57 min.
Better than the current scraper overall	Better on Bray and Cimberio, level on Crane. On Sioux Chief it's mixed: ahead on CAD, images and sizes, behind on main docs and image accuracy.

Speed and requests (first runs, nothing saved)

Site	New step: requests / time	Current scraper: requests / time
Bray	1,227 / 26 min	1,110 / 22 min
Cimberio	404 / 6 min	82 / 1.5 min
Sioux Chief	20,639 / 4 h 09 min	14,408 / 4 h 21 min
Dixon, Apollo	blocked after 8 requests; 0 families	–

With bare CSVs the new step has to download pages and files just to find out what they are. On Bray, about 1,100 of its 1,227 requests were downloads made only to learn a file's type, size or image dimensions. Richer old-agent output removes most of that.

The unsure queue: what the rules couldn't decide

Question type	Bray	Crane	Cimberio	Sioux Chief	Default action
Does this file belong to this family?	265	82	0	386	keep
Page and PDF disagree on a spec	254	130	8	663	keep both, flag
Which document type?	70	19	0	566	keep as spec sheet
Which spec value?	20	5	0	7	keep first-ranked source
Same family?	0	26	10	769 → 497*	keep separate

*Rose to 1,648 after the listing-rule fix, then fell to 497 after the item-table rule.

The AI helper test, in detail

We used Claude (Sonnet, through a subscription login) on Sioux Chief's 497 "same family?" questions. It answered 323 before the usage limit, and we stopped there.

- **Verified "different": 256, of which 83% were right. Verified "same": 59, of which 58% were right.** Mostly it merged spare-parts pages with the product they serve. The quotes were real, so the quote check couldn't catch that kind of judgement error.
- **Why it can't close the gap:** each answer merges at most two families, and most duplicate families never become a question at all. A perfect run would still leave about 4,100 families against 700.
- **Cost:** free on the subscription; about $13 for 497 questions at API prices.

9. Full numbers by site

Each row is one version, measured against the same answer key. The top two rows are the original data: what the old agent's list contains, and what the current scraper produces. The last row is the latest version. Bold rows are the original data and the latest version.

Sioux Chief

% of the answer key

One web page per part number, so it is the hardest site for grouping. *The first live run used older code, and part of its higher numbers came from loose matches on series code alone. "End of rounds" is the real starting point.

Version	Families found	Families correct	Families produced (key: 700)	Duplicates score	Main docs	Other docs	CAD	Images found	Images correct	Sizes found	Specs correct	Specs complete
Old agent link list original data	100%	–	–	–	100%	100%	100%	100%	–	–	–	–
Current scraper original tool	93%	100%	6,324	0.10	46%	90%	14%	14%	79%	3%	15%	29%
First version live run, older code*	59%	100%	3,304	–	36%	17%	86%	37%	50%	9%	20%	37%
End of rounds re-run from saved copies	54%	100%	3,749	–	43%	33%	71%	22%	46%	6%	17%	34%
Richer link list (L1) simulated	54%	100%	3,749	–	43%	33%	71%	21%	33%	6%	17%	34%
Listing-rule fix (Option A) kept	88%	100%	5,725	0.11	43%	87%	71%	27%	50%	4%	18%	35%
Item-table grouping rule in PR	90%	100%	4,574	0.14	39%	87%	57%	40%	61%	19%	16%	32%
Claude helper (323 of 497 questions) stopped	90%	100%	4,515	0.14	39%	87%	57%	40%	62%	20%	16%	32%

Bray

% of the answer key

Almost nothing changed after round 2. The remaining misses are two families dropped as listing pages and two combined product photos.

Version	Families found	Families correct	Families produced (key: 103)	Duplicates score	Main docs	Other docs	CAD	Images found	Images correct	Sizes found	Specs correct	Specs complete
Old agent link list original data	100%	–	–	–	100%	100%	100%	98%	–	–	–	–
Current scraper original tool	97%	82%	122	1.00	93%	95%	100%	80%	85%	100%	67%	86%
First version live run	97%	94%	107	–	85%	81%	100%	78%	84%	100%	67%	85%
Round 1 kept	97%	94%	107	–	85%	81%	100%	73%	86%	100%	67%	85%
Round 2 kept	97%	94%	107	–	93%	95%	100%	73%	86%	100%	67%	86%
Round 3 rejected	97%	94%	107	–	93%	95%	100%	73%	86%	100%	67%	86%
Round 4 + Crane fix kept (end of rounds)	97%	94%	107	–	93%	95%	100%	73%	86%	100%	67%	86%
Richer link list (L1) simulated	97%	94%	107	–	93%	95%	100%	73%	83%	100%	67%	86%
Listing-rule fix (Option A) kept	97%	94%	107	1.00	93%	95%	100%	73%	86%	100%	67%	86%
Item-table grouping rule in PR	97%	94%	107	1.00	93%	95%	100%	73%	86%	100%	67%	86%

Crane

% of the answer key

The Crane fix stopped wrong merges (for example DA931 with DA941), taking families from 208 to 218 against 217 in the key. Round 3 raised "other docs" but made specs worse, so it was undone.

Version	Families found	Families correct	Families produced (key: 217)	Duplicates score	Main docs	Other docs	CAD	Images found	Images correct	Sizes found	Specs correct	Specs complete
Old agent link list original data	99%	–	–	–	100%	100%	67%	100%	–	–	–	–
Current scraper original tool	100%	100%	218	0.99	100%	71%	100%	78%	100%	74%	62%	83%
First version live run	100%	100%	208	–	100%	55%	100%	85%	100%	74%	62%	83%
Round 1 kept	100%	100%	208	–	100%	45%	100%	85%	100%	74%	61%	79%
Round 2 kept	100%	100%	208	–	100%	55%	100%	85%	100%	74%	62%	83%
Round 3 rejected	100%	100%	208	–	100%	94%	100%	85%	100%	74%	62%	83%
Round 4 + Crane fix kept (end of rounds)	100%	100%	218	–	100%	55%	100%	93%	100%	74%	62%	83%
Richer link list (L1) simulated	100%	100%	218	–	100%	55%	100%	93%	100%	74%	62%	83%
Listing-rule fix (Option A) kept	100%	100%	218	0.99	100%	55%	100%	93%	100%	74%	62%	83%
Item-table grouping rule in PR	100%	100%	218	0.99	100%	55%	100%	93%	100%	74%	62%	83%

Cimberio

% of the answer key

The old agent's list is missing about half the main documents and the price list, so 54% is the ceiling for main docs whatever we do.

Version	Families found	Families correct	Families produced (key: 32)	Duplicates score	Main docs	Other docs	CAD	Images found	Images correct	Sizes found	Specs correct	Specs complete
Old agent link list original data	100%	–	–	–	54%	100%	0%	100%	–	–	–	–
Current scraper original tool	97%	66%	47	1.00	25%	12%	67%	0%	–	0%	34%	43%
First version live run	100%	100%	23	–	54%	100%	100%	100%	15%	0%	53%	72%
Round 1 kept	100%	100%	32	–	54%	100%	100%	100%	50%	0%	53%	72%
Round 2 kept	100%	100%	32	–	54%	100%	100%	100%	50%	0%	53%	72%
Round 3 rejected	100%	100%	32	–	54%	100%	100%	100%	50%	0%	53%	72%
Round 4 + Crane fix kept (end of rounds)	100%	100%	32	–	54%	100%	100%	100%	50%	0%	53%	72%
Richer link list (L1) simulated	100%	100%	32	–	54%	100%	100%	100%	50%	0%	53%	72%
Listing-rule fix (Option A) kept	100%	100%	32	1.00	54%	100%	100%	100%	50%	0%	53%	72%
Item-table grouping rule in PR	100%	100%	32	1.00	54%	100%	100%	100%	50%	0%	53%	72%

10. Proposals

10a. Site profile: "how is this site organised?" (built, first results measured)

What it is. A new first step in processing. It runs *after* the old agent and *before* grouping: old agent collects → site profile → grouping and sorting → scoring. It needs nothing new from the old agent: it reads the existing link list and saved page copies, with no new crawling.

How it works. It groups the links by URL pattern and opens 5 pages of each pattern. It looks for a repeated "card" layout (many small boxes, each with a name, a link and a part number). Then it tests five family shapes on about 50 sampled product pages:

- **Category cards:** each card on a category page is a family.
- **Family pages:** one page lists several part numbers.
- **Item tables:** a page lists its sibling part numbers.
- **Shared spec sheet:** a hint only, because one sheet can cover several families.
- **One page per model.**

Each shape is scored on how many pages it explains and how often it contradicts itself. The winner is written to a short readable file, `<site>.profile.json`, with the reason and example pages. A person can check or correct it by hand. It takes 3–15 seconds per site. The answer keys are never used to build it.

Site	Shape picked	Evidence from the sample
Sioux Chief	Category cards	Cards name the family of 100% of sampled pages, with no contradictions. 164 category pages read (out of about 7,000 pages); 698 card families.
Bray	Family pages	88% of sampled pages list several part numbers
Crane	One page per model	95% of sampled pages are one model each
Cimberio	One page per model	91% of sampled pages are one model each

Grouping follows the profile. Pages the profile covers go into its families first. Today's rules only handle the rest, and they can never merge two families the profile keeps apart. On Bray, Crane and Cimberio, today's rules already matched the site's shape, so their scores are **exactly unchanged**.

Sioux Chief results (% of the answer key; "correct" = share of what we picked that the key agrees with):

	Before (item-table rule)	Profile, first version	Profile, fixed
Families produced (real: 700)	4,574	572	**603**
Duplicates score (1 = none)	0.14	0.93	**0.93**
Families found	90	89	89
Sizes/part numbers found / correct	19 / 90	83 / 40	**83** / 48
CAD found / correct	57 / 100	100 / 78	**100 / 100**
Main docs found / correct	39 / 92	46 / 33	46 / **46**
Other docs found / correct	87 / 96	87 / 84	87 / 90
Images found / correct	40 / 61	47 / 46	51 / 52
Specs / completeness	16 / 32	19 / 36	20 / 36
Price info right	19	83	83
Processing time (saved copies)	52 min	~14 min	27 min

- **The big win:** the "6× too many families" problem is mostly solved: 4,574 → 603 against 700 real ones. Sizes, CAD and price info jump with it.
- **The fix between versions:** parent category pages repeat their sub-categories' cards under the same names, and the first version merged them. Now a part number listed on several category pages belongs to the most specific one (the page with the fewest cards). That is a general rule about category sites. We found it by comparing with the answer key, and we say so in the report.
- **What still fails:** on a few cards, the site lists several product lines under one name, which the answer key splits by hand. For example, one card covers 6 families in the key. A merged family picks up the other families' documents and images. Two of the 15 hand-checked families are like this, so "correct" main docs fall from 92% to 46%. By our keep rule (no key number may drop more than 2 points), it doesn't pass yet. So the profile is **opt-in** (`process --profile`), not the default.
- **Why some profile families are still mixed:**
  - One card shows the same product across several pipe types, while the key splits by pipe type.
  - The card names a size or connection; the real family name (the series) sits above the card.
  - A small card includes related items, such as a tool or accessory, that the key treats as their own family.

Code and full numbers: PR #7, `scraper/mfr_scraper/process/profile.py`, REPORT_APPROACH_B.md section 14. 85 unit tests pass.

Proposed next: a shared shape catalogue (written, not yet built or tested)

The goal is a profile step that is repeatable and accurate on new manufacturers, with no rules written for a single site.

1. **Rules live in one harness document** (`SITE_PROFILE.md`, next to SPEC.md). It lists a small, fixed set of general family shapes: category cards, family pages, item tables, one page per model. It also lists building blocks, such as "series heading above the cards" or "pipe type in the category or product name". For each one it states how to detect it, its pass bar (coverage and contradictions), how families are formed, and its unit tests.
2. **A site profile holds settings only, never logic.** For example: shape = category cards; card = this box; family name = series heading + card name. The same code runs for every site.
3. **Code tries first.** If a catalogue shape passes its bar, code builds the profile in seconds, for $0, with no AI.
4. **New, unrecognised sites go to the site profile agent** (below), which may only choose from the catalogue.
5. **The catalogue changes only by a reviewed PR.** A new shape must be described in general terms, work on at least 2 manufacturers (including one it wasn't built on), and make no existing site worse (keep rule).
6. **Proof is blind.** We build answer keys for 3–5 new manufacturers, run with no tuning, and report the scores as they are.

Site profile agent (design only, not built)

Code handles every site it recognises. The agent is only for sites where no catalogue shape passes its bar.

Question	Design
When it runs	Only when code finds no catalogue shape that passes, or the winning shape has too many contradictions. Off by default.
What it sees	The catalogue (SITE_PROFILE.md), 10–20 sample pages from the old agent's list (saved copies), and code's score for each shape. Never the answer key.
What it returns	A shape and its settings, chosen from the catalogue. For example, which page element is a card and where the family name sits. Each setting is backed by a word-for-word quote from a sample page.
What it may not do	Invent a rule, write code, or list families by hand.
How its answer is checked	Code confirms the shape is in the catalogue and every quote appears in the pages. It then rebuilds the families with those settings and scores them on coverage and contradictions. **Pass:** the profile is saved, marked "decided by agent". **Fail:** it's marked "needs review" and the standard rules are used.
If nothing in the catalogue fits	The agent says so and describes the pattern. A person decides whether it becomes a new general shape: a reviewed PR that works on 2+ manufacturers and passes the keep rule.
Repeatability	Accepted settings are saved in the profile. Later runs reuse them with no new AI call, so the same input gives the same output.
Cost	A few AI calls per new, unrecognised site (estimate); $0 for recognised sites.

Step	What	Expected effect (estimate, untested)	Effort
a	Load the 103 missing category pages; confirm the pipe-type and series clues are on the pages	Shows whether the first two causes can be fixed from site data	~15 min
b	Write the catalogue, move profiles to settings only, add the two building blocks, send small mixed cards to unsure	Families mixed by the first two causes split correctly; aim: Sioux Chief passes the keep rule	2–3 days
c	Answer keys for 3–5 new manufacturers + blind run	The real test of "works on new sites"	~1 session per site

10b. Free extra fields from the old agent

The rule is no new requests: the agent records only what it already has in hand while crawling. Output is one JSON line per page or link (the format and validator already exist).

Priority	Fields	What it unlocks
1. Relationships	found_on (page the link was on), link text, section heading, in header/footer/menu, link kind	Which docs and images belong to which product, without crawling again
2. HTTP facts	status, final URL, content type, size	Dead links and true file types, without downloading
3. Page facts	**saved page copy**, title, breadcrumb, canonical URL, language, structured product data, has spec table	Works on sites that block us (Dixon, Apollo), removes page fetching (68% of Bray's run time), and gives cleaner grouping
4. Image details	alt text, width/height, srcset, plus a content fingerprint	Picks the best image without downloading every size. Without the fingerprint, duplicates slip through (measured: −3 to −13 points of image accuracy).
5. Change detection	etag, last-modified, sitemap date, crawl time, agent version	Cheap re-runs: only changed items are re-checked

Also: one fixed label set for `url_type`, normalised URLs, and "tag, don't drop": language copies, footer links and off-site links are kept with tags. Acceptance checks: every link's "found on" page exists in the file; run time stays within ±5%; the file passes the validator. Separately, the agent's discovery gaps (Cimberio's price list and documents) need fixing on its side, because no extra field fixes them.

10c. Better specs (the biggest gap on every site)

- A shared dictionary of spec names, e.g. "Working Press." means pressure rating.
- Fewer false page-vs-PDF conflicts. Most flagged conflicts are formatting differences, e.g. "-10 to 100" vs "10 - 120°C".
- AI reads spec tables and PDFs, and every value must come with a verified quote. This is where AI is most useful: an estimated +2–3 average points from conflicts alone.

Effort: about 1 week.

10d. Smaller rule fixes already identified (no new data needed)

- Match "non-product" URL words only as whole path segments. This rescues Bray's "series-70-hazardous-location", for example.
- Let spec tables and spec downloads outweigh "listing page" cues for pages without a product word in the URL. This would cover Bray's 2 remaining dropped families.
- Name combined product photos correctly (Bray images).
- Keep multi-product documents without creating false spec conflicts (Crane other docs: 55% vs 71%).
- Keep fingerprinting images the step decides to keep, even when their size is already known.

10e. Prove it on new manufacturers

Build answer keys for 3–5 varied manufacturers, about one session each, using the same process. Include electrical, plumbing and mechanical makers, one with assets on a CDN and one with part-number pages. Then run blind, with no tuning. Milwaukee and Caleffi stay reserved for the final blind test. This is the only honest way to say "it works for all manufacturers".

10f. Database for production (DATA_MODEL.md, design only)

Three services share Postgres tables and never call each other:

- a **link collector** (the old agent);
- a **processor** (this step);
- a **health checker**, which re-checks links and queues the families that changed.

Tables	Why
sites, links, **page_links** (which page links to which), link_versions (each stored copy with its fingerprint)	The relationships are the main grouping evidence. Every value traces back to the exact copy it came from.
families, family_assets (with keep/drop/unsure + reason), variants, spec_values + spec_sources, prices	The output plus its audit trail. Page and PDF values are both kept; prices sit on the variant, or on the family when it has no variants.
runs; plus ai_decisions and rules for the self-healing loop	Runs can be compared and rolled back. Only new or changed links are reprocessed, which is essential at thousands of sites.

Evaluation stays file-based, through an exporter, so the answer keys keep working. Migration: loader → storage interface → old agent writes to the tables → schedule the services.

11. Plan, decisions and risks

Step	Work	Success check	Effort
1	Site-profile step (10a). **Built:** 3 sites unchanged; Sioux Chief 4,574 → 603 families. Still to do: split mixed cards, or a team decision on the key.	No regressions on 3 sites, and no key number on Sioux Chief drops more than 2 points	Done
1b	Shape catalogue (10a): rules in SITE_PROFILE.md, profiles as settings only; then the site profile agent for unrecognised sites (chooses from the catalogue only)	Sioux Chief passes the keep rule; no site gets worse; new shapes only via reviewed PR, proven on 2+ sites	2–3 days, then 2–3 days for the agent (estimate)
2	Old-agent fields (10b), starting with found_on and page copies	Validator passes; run time ±5%; Dixon/Apollo produce families	On the old agent's side
3	Specs (10c) and rule fixes (10d)	Spec accuracy and conflict precision up ≥ 5 points	~1 week
4	New answer keys + blind test (10e)	Scores hold on unseen manufacturers with no tuning	~1 session per site
5	Self-healing loop, then database (7, 10f)	Unsure items and AI cost fall run over run; no score drops	After steps 1–4

Honest status against 97%: we are not there. Families on page-per-family sites (Bray, Crane, Cimberio) are at 93–100%. On Sioux Chief, the site profile brings families to 603 (700 real), but main-document accuracy is still too low. Specs (15–67%) are far off everywhere. Steps 1–2 target grouping and documents, step 3 targets specs, and step 4 is the proof.

Decisions the team needs to make:

1. Can we get the old agent's code, or ask its owners for the extra fields?
2. For the AI layer: which model, what cost per site is acceptable, and should learned rules apply automatically or go to review first?
3. Is some run-to-run variation acceptable on the unsure items? It can be reduced with fixed model versions and cached answers.
4. What is the speed target for a 6,000-product catalogue?
5. Which manufacturers get answer keys next?
6. Should spare-parts lines count as separate families (as the keys do now) or join their product?
7. Who reviews and approves new shapes in the catalogue, and is a person needed to confirm each new site's profile at first?
8. When a site's own category card covers several product lines, is that one family (as the site shows it) or several (as the key splits it)?

Risks:

- Collecting wider means more for the sorter to get wrong. This is why there's a precision gate.
- AI answers can be confidently wrong, even with real quotes, as the 58% result showed. Use AI for reading, not for structural decisions.
- Blocked sites can't be scored.
- Results on only a few sites misled us once already.

Appendix: where everything lives

Repository: github.com/ved-dave/manufacturer-scraper-eval. The Approach B work is in PR #7 (not merged).

Document	Contents
SPEC.md	The rulebook
harness/ (gold/, mfr_harness/)	Answer keys, ambiguity logs, blind re-label notes, validator and scorer
REPORT.md	The current scraper: 10 rounds, blind test, gap analysis
scraping-approach-proposal.md	"Collect wide, sort smart" design and the self-healing loop
OLD_AGENT_IMPROVEMENTS.md	Field-by-field request for the old agent
DATA_MODEL.md	Production database design
REPORT_APPROACH_B.md, AUTONOMY_LOG.md	Every Approach B experiment, with numbers and decisions

Sources: manufacturer-scraper-eval repository (SPEC.md, REPORT.md, REPORT_APPROACH_B.md, scraping-approach-proposal.md, OLD_AGENT_IMPROVEMENTS.md, DATA_MODEL.md, harness score files). All scores use the unchanged answer keys.